3000-L Scale Perfusion Demonstration and its Implications to End-to-End Continuous Downstream Process Design

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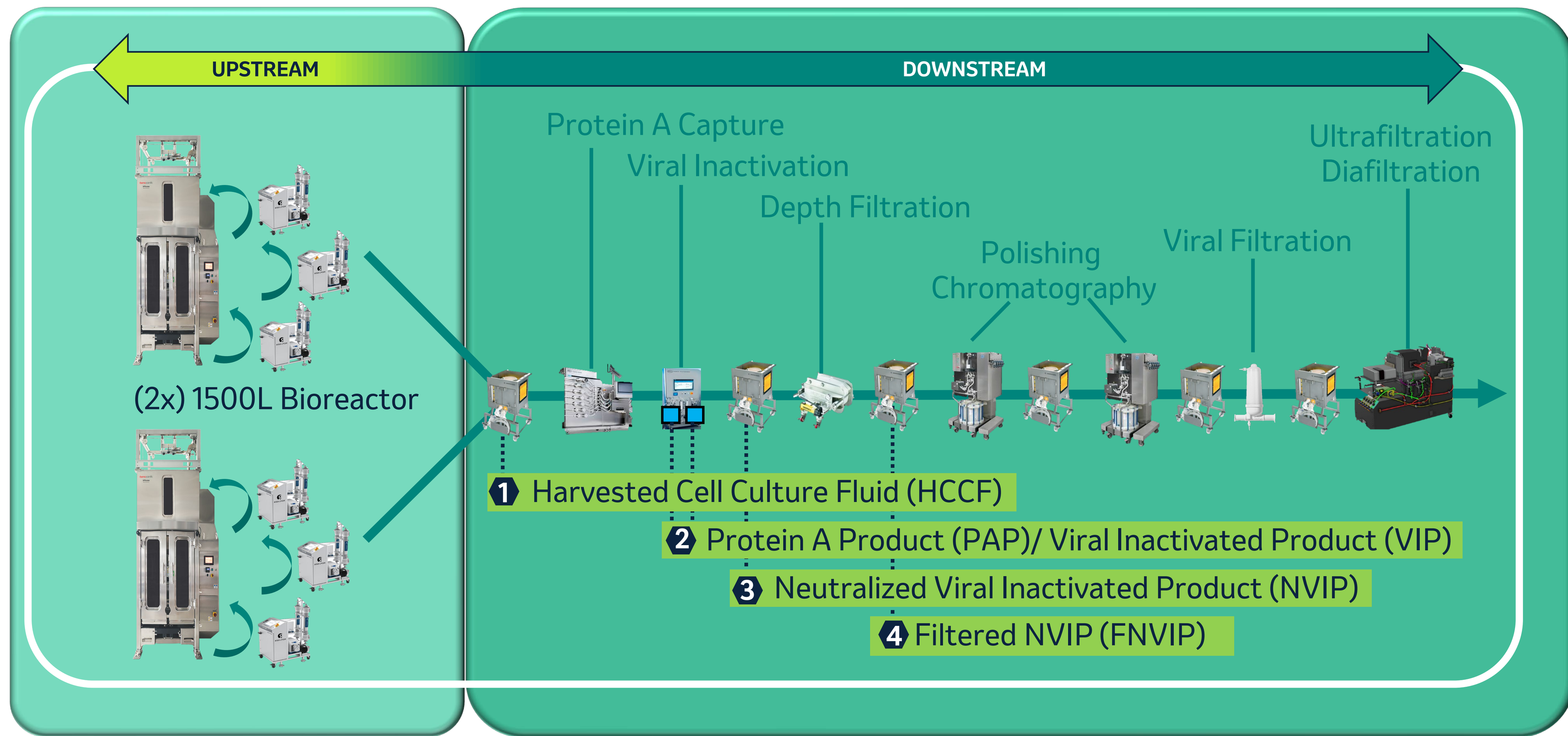
After a decade of process development efforts in Continuous Manufacturing (CM) of therapeutic proteins, commercial production is within immediate reach. This presentation will not only focus on Merck's intended scale for end-to-end (E2E) commercial process with a demonstration of 3000-L perfusion cell culture, but also the design of downstream surge vessels to support the corresponding mass flowrates.

To establish the perfusion cell culture strategy for commercial scale, an intermediate 500-L clinical scale was used to support scale-up efforts and define the clinical scale. A successful scale-up demonstration of 500-L to 3000-L of cell culture bioreactor will be shown, in addition to key decisions on technology chosen, namely, two 3000-L N-stage perfusion bioreactors at half full volume and scale-out of cell retention devices (while using same seed train).

In order to equip the E2E downstream for commercial scale mass flowrates, this presentation will also address surge vessel design based on process dynamics understanding. The basis of design will cover concepts of residence time distribution and the impact of intermediate stability to identify the commercial platform vessel size. The final intent is to demonstrate Merck's readiness for GMP continuous process in order to meet commercial demands.

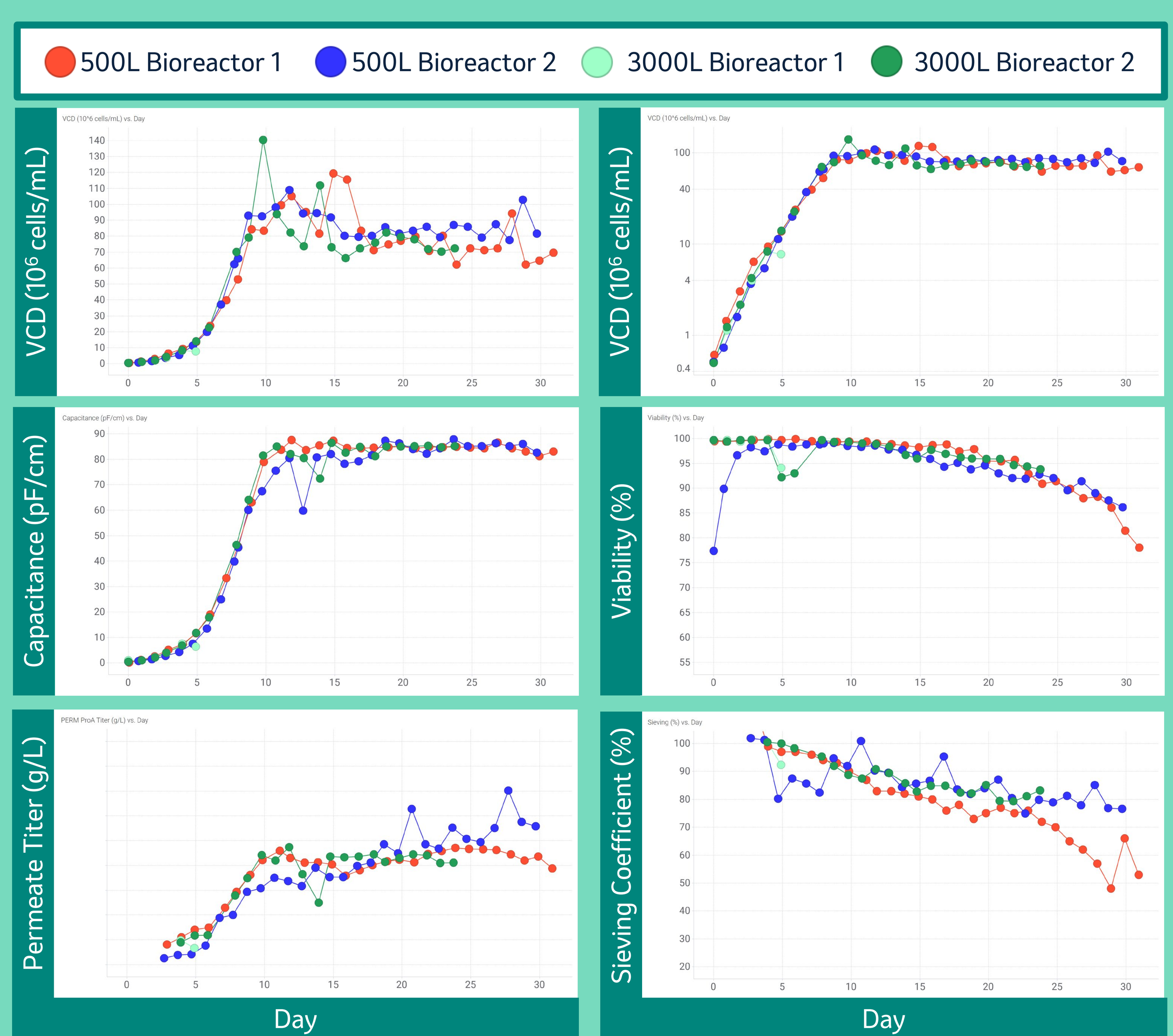
3000-L End-to-End Continuous Manufacturing Commercial Scale



3000-L Perfusion Cell Culture Demonstration

Monoclonal Antibody Perfusion process

- Host GS-KO CHO, in-house perfusion media, 2 VVD exchange rate
- Capacitance-based bleed control
- Working volume 500-L perfusion cell cultures:
  - Bioreactor 1 – 90%; Bioreactor 2 – 100%
- Working volume for both 3000-L bioreactors was 50% (1500L)
  - N-1 batch 500-L bioreactor was used to inoculate both tanks
  - Each bioreactor had (6x) 8.8 m² Microza TFF filters inline (and 6x backups)



Large volume perfusion cell culture scale-up strategy

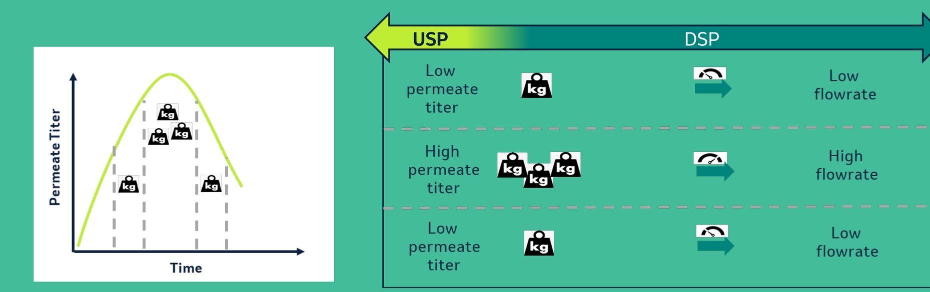
- Bioreactor scale-up based on maintaining constant Power/Volume and gas flowrate (vvm)
  - Ensuring sufficient oxygen mass transfer and adequate carbon dioxide stripping
- Cell retention device scaling out
  - Maintaining shear stress and tangential flow filtration parameters (cross flowrate, permeate flux)

Discussion

- Successful scale-up demonstration from 500-L to 3000-L perfusion cell culture
  - Consistent: Cell growth; state of control cell density and viability; permeate titer
- Data shown is from the first 3000-L runs, where the intent was to demonstrate consistent growth profile for both bioreactors from the same N-1 inoculum (3000-L Bioreactor 1 stopped on day 5 of cell culture – light green)
- Initial drop in viability on 500-L bioreactor 2 (teal) was due to an intentional exposure to high agitation rate to evaluate cell line sensitivity (P/V at 60 W/m³, then lowered to 30 W/m³)
- TFF filter swap was performed on 500-L bioreactor 1 (red) on culture day 29
- High technical complexity in setup and operation of multiple cell retention devices

Downstream Intermediates Stability in the Context of Continuous Manufacturing

Fluctuation in permeate titer impacts downstream operation and flowrate setpoints



Hypothetical HCCF Titer Setpoints for GMP

HCCF Titer (g/L)		
Setpoint[a]	Min Value	Max Value
0.8	0.6	0.99
1.2	1.0	1.29
1.4	1.3	1.49
1.6	1.5	1.79
2.0	1.8	2.30

- Requires to pause continuous process and make changes according to new mass loading
- Changes to downstream flowrate impacts, product residence time in surge vessels

Pause limit assessment for Continuous Manufacturing intermediate process locations

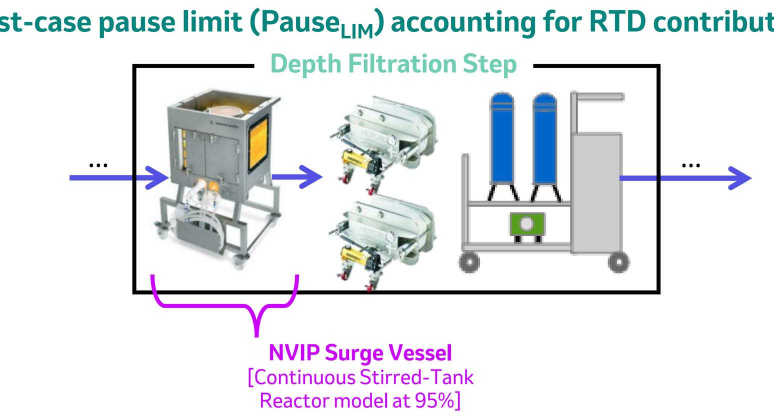
1. HCCF surge vessel – No CM relevant pause limit, HCCF flows continuously either to waste or ProA step
2. VI tanks – Pause limit for batch VI leverages intermediate hold times directly from process description
3. NVIP surge vessel – Pause limit needs to account for highest allowable fill volume of 90%
4. FNVIP surge vessel – Pause limit needs to account for highest allowable fill volume of 60%

Contextualizing batch intermediate biochemical stability experiments in Continuous Manufacturing

Typical batch intermediate stability limits from process description

Process Intermediate[b]	Allowable hold time at room temperature (15-25°C)	Allowable hold time limit at 2-8°C
HCCF [a]	≤ 3 days	≤ 10 days
PAP	≤ 3 days	≤ 3 days
VIP	≤ 120 minutes	N/A
NVIP	≤ 3 days	≤ 3 days
FNVIP	≤ 3 days	≤ 3 days
MMAEXP or CEXL	≤ 14 days	≤ 14 days
CEXP	≤ 14 days	≤ 14 days
VFP	≤ 14 days	≤ 14 days

Worst-case pause limit ($Pause_{LIM}$) accounting for RTD contribution



$$Pause_{LIM} = t_{Hold\ Time} - \frac{3 \cdot V_{Surge\ Vessel}^{High-high}}{Q_{Slow}}$$

Define posed limits in CM for NVIP surge vessel at 90% fill volume

Parameter	Unit	Attribute
Batch intermediate holding time ($t_{Hold\ Time}$)	h	72[a]
Highest Surge Vessel Volume ($V_{SV}^{High-high}$)	L	45[b]

Process dependent inputs: Pause limit is dependent on permeate titer

Parameter	Unit	Attribute				
Permeate titer[b]	**g/L**	**0.8**	**1.2**	**1.4**	**1.6**	**2**
Low Flowrate (Q_{Low})[b]	mL/min	39	47	55	63	79
Surge Vessel RTD	h	57	48	41	36	29
Pause Limit ($Pause_{LIM}$)	**h**	**14.8**	**24.3**	**31.1**	**36.2**	**43.4**

Discussion

- Operating within pause times limits based on batch biochemical stability data ensures consistent product quality
- Intermediates residence time in surge vessels is dependent on allowable highest fill volumes and lowest flowrates
- Downstream flowrates change proportionally to permeate titer with significant impacts on pause times (hypothetical example range from 14 to 43 hours)

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